When to Upgrade Your CPU: Evidence-Based Recommendations from Performance Equivalence Charts

César H. R. Soares, Getúlio Mendes, Mel Raposeiras Ricaldoni, Nestor D. O. Volpini

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Abstract

Since the establishment of the Von Neumann architecture, computer processors have undergone remarkable performance improvements driven by successive design innovations. This paper presents a performance-oriented analysis of major processor generations. Using benchmark data and technical reviews, we examine how disruptive innovations have affected single-core and multi-core performance, highlighting periods of accelerated progress and relative stagnation. By framing processor evolution through performance trends, the results provide insights into performance dynamics and equivalence charts that support more informed CPU upgrade and technological investment decisions, with implications for both economic and ecological sustainability.

1 Introduction

With the constant advancement of technology, processor manufacturing processes have reached unprecedented levels of efficiency, leading to their widespread integration into consumer goods and electronic devices. The advances were not limited to the production scale but also extended to hardware performance. This was notably formulated in Moore's law, which states that the number of transistors that can be integrated into a chip doubles approximately every two years, resulting in exponential growth in performance [Mac11].

With this increase in industrial productivity, driven by the desire to maintain or increase profit margins and avoid the rapid obsolescence of products, companies significantly shortened the intervals between product generations, resulting in shortened market cycles for these items and not always the same level of improvement in CPUs in every cycle [LPVZ16]; further complicating the decision-making process.

As market cycles are shortened, a complexity arises in the task of acquiring a new processor or, eventually, a computer. The market is flooded with a vast number of options, making the choice especially challenging for the average user. This scenario is aggravated by the "grey" market, which adds older processors that ceased being fabricated into the equation, further complicating matters for the lay consumer, who may not understand the cost-to-performance ratio of what they are acquiring.

The lack of clarity and accessible information regarding the cost-performance of models hinders decision-making, since, in most cases, users of computing systems do not possess in-depth technical knowledge about the internal components of their devices.

This scenario often results in unnecessary processor upgrades, driven by the mistaken perception of gains, which may be not significant in terms of performance. This leaves consumers particularly vulnerable to making decisions based on marketing campaigns or suggestions from unreliable and non-specialist sources [SH09].

Therefore, the fundamental purpose of this paper is to present a methodology that unfolds into a guide, detailing the cost, performance, and relevance relationships of CPUs, with a focus on the end consumer. For this, an analysis of data provided by PassMark, generated from the CPUmark software, allowed for a better understanding of the lifespan of a chip, that is, how long it can remain relevant in the market. An assessment that enables a global view considered as criteria performance, price, energy efficiency, and functional innovations incorporated into the product.

This methodology can later be adapted to use other data generated by other benchmarks, thus representing a significant step toward a generalist way of analyzing the appropriate information to evaluate the timeliness of upgrading or acquiring CPUs or technological infrastructures.

2 Theoretical foundation

For many years, Intel adopted the “Tick-Tock” model as a strategy for the development of its processors [Int14]. This model divided the development cycle into two distinct phases. The “Tick” microarchitectures represented a significant advancement in terms of design and functionalities, introducing new manufacturing technologies (such as the transition to a smaller node, for example, from 32nm to 22nm) and substantial architectural changes. A “Tick” generally involved the implementation of a new manufacturing process, as exemplified above, resulting in better performance and lower energy consumption, eventually enabling higher clock speeds, thereby slightly increasing performance.

The “Tock” microarchitectures, in turn, functioned as a refinement of the “Ticks”, focusing on performance optimization and, frequently, lithography reduction, without necessarily introducing disruptive new technologies in the processor core design. Instead, a “Tock” concentrated on optimizing the existing microarchitecture, increasing clock frequencies, and improving energy efficiency. This allowed Intel to maximize the return on investment in new manufacturing technologies and core architectures.

A notable example of this pattern was the Skylake microarchitecture, launched by Intel in 2015. Skylake brought with it a series of important innovations, such as support for DDR4 RAM, which provided a significant increase in CPU-to-memory data transfer and overall system performance, especially in applications heavily dependent on memory speed, such as gaming and video editing. Additionally, the Skylake architecture incorporated new instruction sets into the ISA (Instruction Set Architecture) of its processors, expanding processing capabilities and optimizing the execution of various workloads, including instructions for encryption and media processing [Int15]. Skylake also introduced improvements in energy efficiency, allowing laptops and mobile devices to achieve longer battery life [D+17].

Following this, in 2016, Intel released the Kaby Lake microarchitecture, which represented a “refresh” of Skylake. Kaby Lake processors maintained the same number of cores and threads, the same ISA, and a very similar design, but featured an increase in clock frequencies, resulting in slightly superior overall performance compared to the previous generation. The most significant addition of this generation was the introduction of support for 4K UHD content playback with HDCP 2.2 protection.

AMD, Intel’s main competitor, also follows a very similar approach, especially after the launch of processors based on the Zen architecture. A notable example is the Zen microarchitecture and its successor, Zen+. The original Zen represented a milestone for AMD, bringing a new core design and significant improvements compared to the older FX-series processors based on the Bulldozer architecture. Zen introduced processors with a greater number of cores than previously seen in the desktop market, with these cores being much more efficient than the old Bulldozer cores. The Zen architecture also brought significant improvements in performance per clock (IPC), bringing AMD back into competition [SSR+18].

Zen+, in turn, was a refinement of Zen, with optimizations in the manufacturing process (moving from 14nm to 12nm), increased clock frequencies, and other incremental improvements such as enhancements in the Infinity Fabric and memory latency. Zen+ also introduced Precision Boost 2 technology, which allowed AMD processors to dynamically adjust clock frequencies based on workload and temperature conditions, maximizing CPU performance; however, as with an Intel “Tock” architecture, the performance gain was not substantial.

All of this makes the process of choosing the ideal moment and/or the ideal CPU model for a hardware upgrade complex. The difference between a Bulldozer processor and a Zen processor is substantially more significant than the upgrade from a Zen to a Zen+, due to Zen being a new architecture and Zen+ being merely a refresh. Moreover, a high-end model from an older generation, such as an i7 or even an i9, may perform worse than an entry-level model like an i3 depending on the number of generations between them, as in the case of the i7 3770K and the i3 9100.

Thus, the appropriate choice of a processor becomes a complex problem involving the analysis of different brands, models, and generations. It is not reasonable to expect the general public, without specialized knowledge or awareness of the full breadth of market options, to make the best choice for their needs—especially household consumers. As a consequence, in addition to inefficient resource allocation, poor choices may increase the amount of electronic waste, which, being harmful to the environment, is rarely disposed of correctly [Uni22].

3 Methodology

To conduct this study, we initially collected CPU specifications directly from the official websites of each manufacturer, Intel and AMD. Performance data was extracted from CPU Mark, a general-purpose CPU benchmark developed by the company PassMark, which evaluates multiple criteria to compare CPUs [Pas25]. It was chosen for its popularity and readability, making it ideal for lay users seeking to determine which CPU is better between two models, in a general sense.

For this purpose, processors released after Intel's Sandy Bridge microarchitecture and AMD's Zen architecture were selected, with the goal of ensuring the relevance of the analyzed models for current users. AMD processors that preceded the Zen architecture were not considered due to the considerable performance gap between Bulldozer-based and Zen-based processors.

To model the evolution of single-core performance, a linear regression was applied to the historical PassMark CPU data using the least squares method. The linear function was selected for presenting a high coefficient of determination ($R^2 = 0.96$) compared to polynomial and exponential models. The resulting equation:

$$y = 227.18x + 1481.18 \tag{1}$$

where x = processor generation - 1 and y the benchmark score, demonstrates a trend of consistent growth, with an average increase of approximately 227 points per generation.

It's important to note than only the data from Intel's i7 and AMD's ryzen 7 were used for the regression analysis, this choice was made in order to capture the high-end of processors development that is more representative of the technological advances made.

In evaluating multi-core performance, a regression analysis revealed an exponential growth pattern, characteristic of the increase in the number of cores and architectural efficiency in recent generations. The exponential function:

$$y = 4865.18 \cdot e^{0.21x} \tag{2}$$

was selected for presenting the best coefficient of determination ($R^2 = 0.95$), outperforming linear and polynomial models. At this point the interpretation is a bit more complex, the exponent 0.21 indicates an approximate growth rate.

It is important to note that despite the choice of CPU Mark, the method can be adapted for other existing benchmarks. With the data in hand, we developed a Python script to process the information and generate a variety of graphs, including scatter plots and others that assess generational gains of each CPU and the relationship between price and time of use. Special attention was given to equivalence charts, which highlight performance equivalences between low and high-cost models across different generations.

4 Results

4.1 Performance Over Time

In equation 1 we can see that their single-core evolution exhibits behavior similar to a linear function, y represents the single-core performance index, while x denotes the CPU generation. The angular coefficient of 227.18 indicates the rate of increase in single-core performance per generation, while the constant term 1481.18 adjusts the model. This linear relationship suggests that although single-core performance continues to improve in a steady and predictable manner, it's important to note that this linear progression may reflect physical and architectural limitations, such as the difficulty of increasing clock frequency without compromising stability and power consumption.

In contrast, multi-core performance displays a more pronounced growth pattern in processors, approaching exponential behavior. This phenomenon is captured by equation 2 where y represents the multi-core performance index, and x retains the same meaning as in the previous equation. The factor 4865.18 scales the initial performance, while the exponent 0.21 determines the rate of growth. The exponential form of this equation reflects that multi-core performance gains are significantly greater compared to single-core. This exponential growth is a direct result of advancements in multi-core chip architecture, where, since the 8th generation of Intel processors, the number of cores has been steadily increasing.

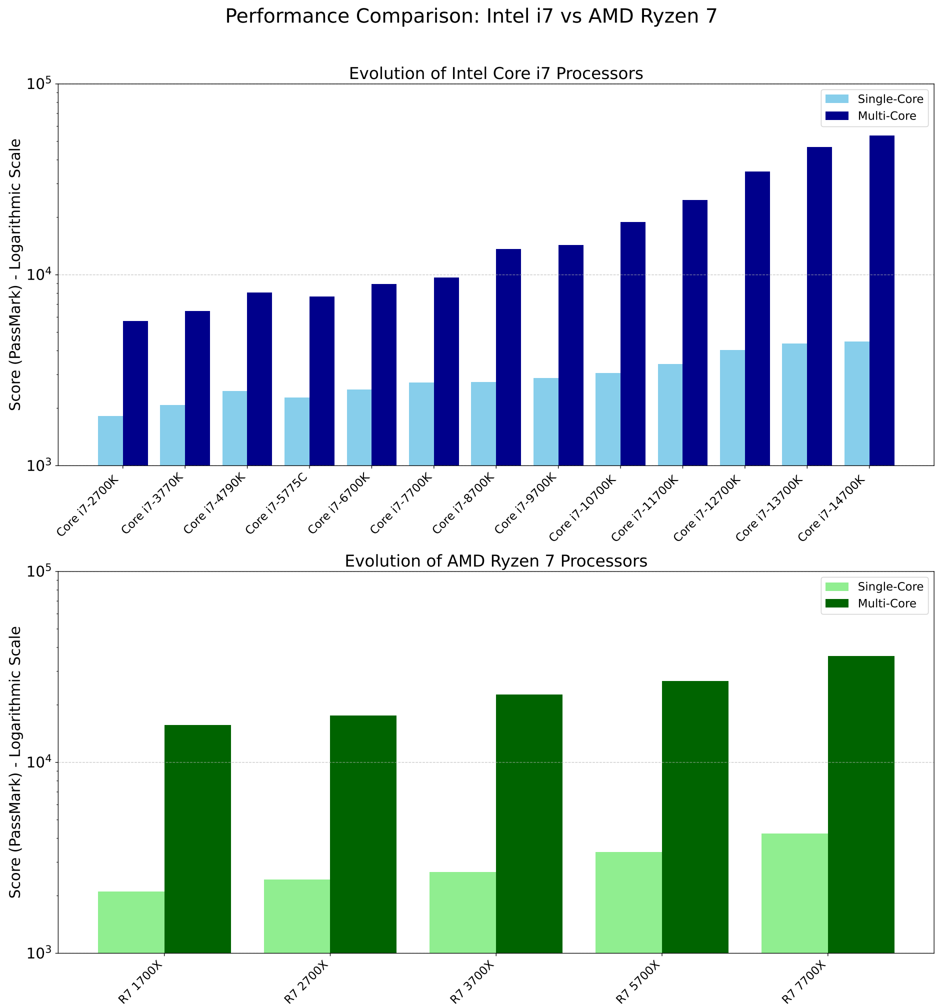

Figure 1: Performance in Single-Core and Multi-Core

Figure 1 illustrates the performance progression of Intel and AMD processors, evaluated from two crucial perspectives: single-core and multi-core. The graph allows visualization of how processing capacity has evolved in both configurations over time.

The difference between the linear and exponential trends highlights a fundamental shift in how processor performance is improved. While single-core performance is limited by factors such as clock speed and improvements in a single core's microarchitecture, multi-core performance can be scaled by adding more cores—which, broadly speaking, facilitates CPU development but complicates software development and optimization to take advantage of parallelism. This shift has important implications for software development, as it encourages programmers to write code that can be executed in parallel in order to fully leverage modern hardware.

In contrast, multi-core performance displays a more pronounced growth pattern in processors released after the 12th generation, approaching exponential behavior. This phenomenon is captured by equation 2 and can be viewed in Figure 2

The Figure 2 illustrates the evolution of processor performance across generations, comparing each model's performance with that of its direct predecessor, in both single-core and multi-core scenarios. This analysis makes it possible to identify the incremental advances brought by each new architecture.

A notable case is that of the i7 5775C, which showed a performance reduction compared to the i7 4790K. This regression is atypical and may be attributed to a combination of factors. First, the i7 5775C operates at a lower clock frequency than the i7 4790K. Additionally, the Broadwell microarchitecture, found in the i7 5775C, introduced an L4 cache (eDRAM) integrated into the processor package, aimed

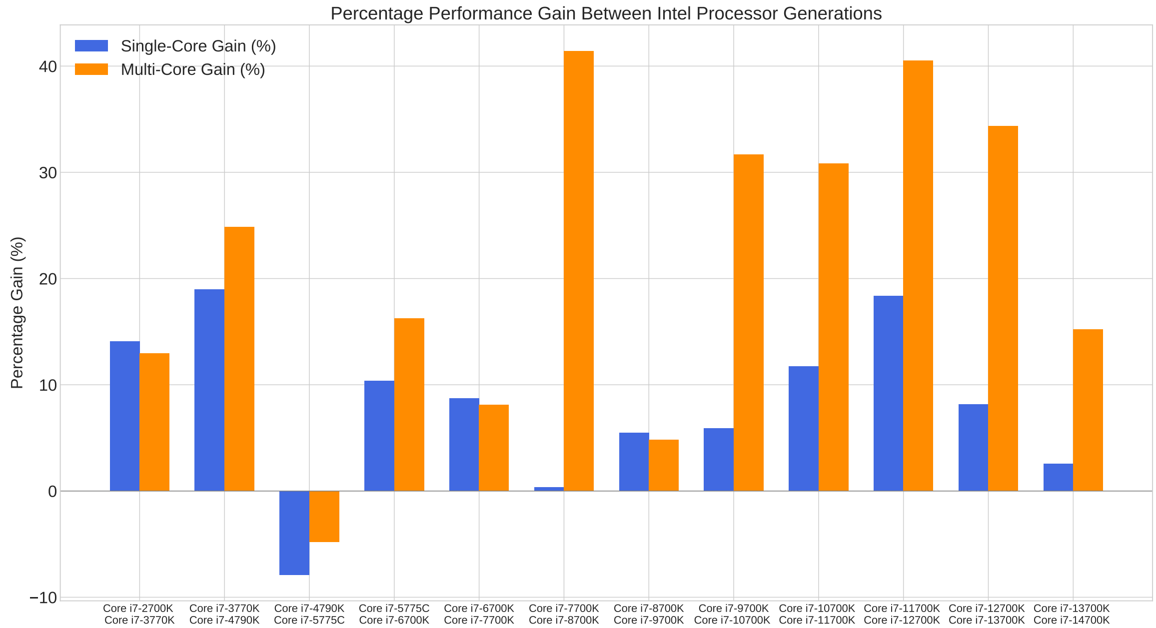

Figure 2: Gain in Percentage, Considering the Previous Generation

at accelerating access to frequently used data. However, if the benchmark tests are not optimized to take advantage of this L4 cache, its potential is not fully realized, resulting in lower-than-expected performance. In other words, the benchmark used may not have been representative of workloads that would benefit from the L4 cache.

The transition from the i7 7700K (Kaby Lake) to the i7 8700K (Coffee Lake) marked a significant turning point, with a 50% increase in the number of physical cores, from four to six. This substantial increase resulted in an approximate 40% gain in multi-core performance. It is important to highlight that the gain is not linearly proportional to the number of cores due to factors such as scheduling overhead and suboptimal inter-core communication. This change demonstrated the growing focus on parallel computing and stands as a milestone in that direction.

Another highlight was the 12th generation of Intel processors (Alder Lake), which introduced an innovative hybrid architecture combining high-performance cores (P-cores) and efficiency cores (E-cores). P-cores are optimized for tasks requiring high responsiveness and performance, while E-cores are designed to maximize energy efficiency and efficiently handle background tasks and parallel workloads. This approach enabled optimized resource allocation, directing the most critical tasks to the P-cores and the less demanding ones to the E-cores, resulting in better overall performance and improved energy efficiency.

On AMD's side, the transition from the Ryzen 7 2700X (Pinnacle Ridge) to the Ryzen 7 3700X (Matisse) stood out especially in multi-core performance. Improvements were driven by a combination of factors, including a significant increase in L3 cache and enhancement of Infinity Fabric, AMD's proprietary interconnect that facilitates communication between CPU cores and RAM.

4.2 Single-Core Equivalence Chart

From the graph in Figure 3, it can be observed, for example, that all models in the Ryzen 1000 series show equivalent performance in single-core, a result consistent with the fact that they share the same initial Zen architecture and are manufactured under the same technological process. This behavior reinforces the relationship between design uniformity and performance consistency in CPUs of the same generation. A noteworthy highlight is the Ryzen 3 1300X, which demonstrates comparable capability to the Intel Core i7 3770K, a high-end processor launched five years earlier. This proximity illustrates how technological advances in mid-range segments can match high-performance solutions from previous generations—a nuance easily overlooked when focusing solely on high-end and entry-level models, but important for making a sound choice.

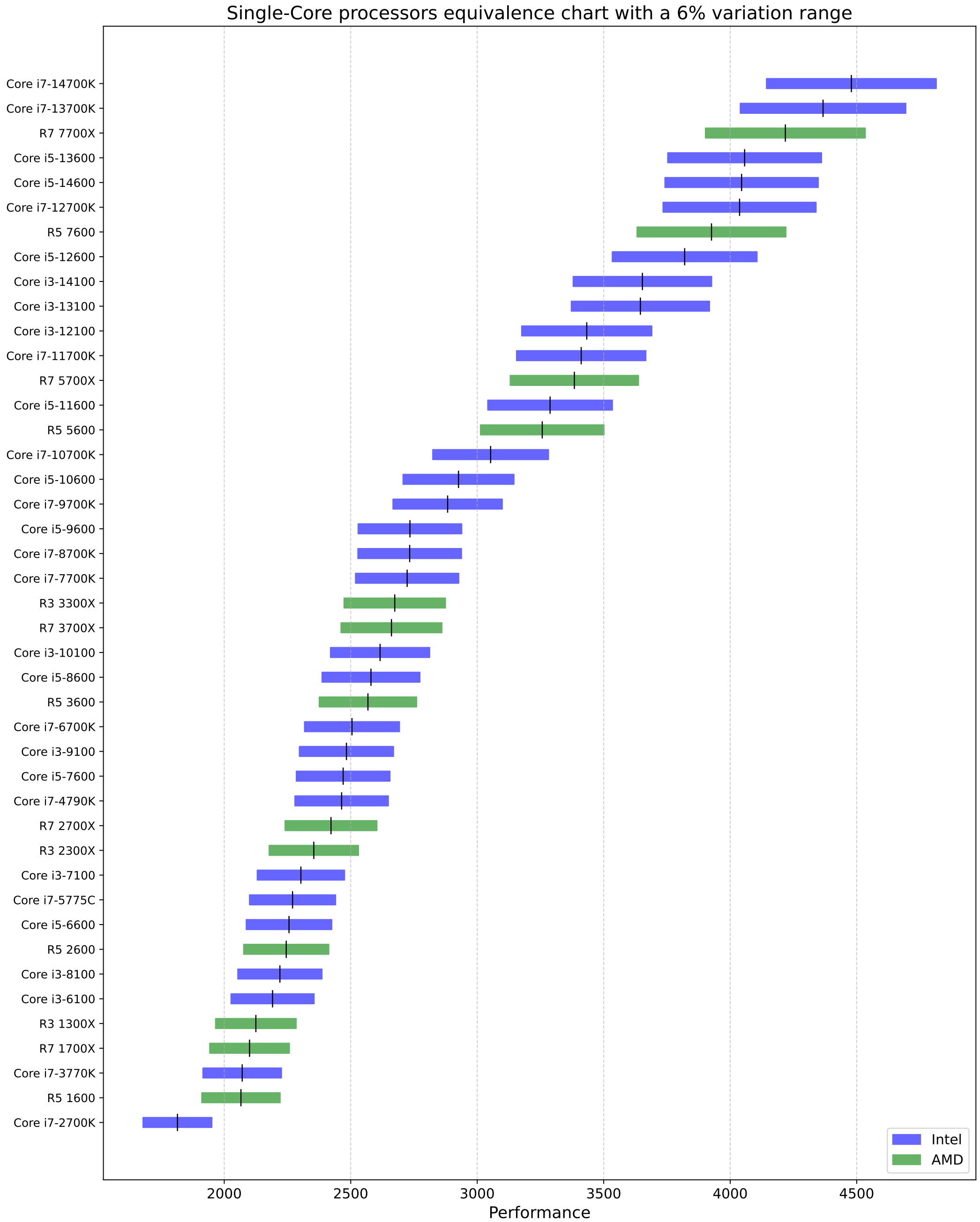

Figure 3: Single core equivalence chart

Examining the evolution of Intel processors, it is noted that Core i3 models from the 6th, 7th, and 8th generations maintained similar performance in single-core, indicating a period of stagnation. However, a significant jump occurs between the 10th and 12th generations, driven by widespread architectural improvements such as the adoption of the 10 nm process and core design optimizations. Core i3 processors from the 12th, 13th, and 14th generations consolidate only very small gains. In parallel, Core i5 models from the 11th, 12th, and 13th generations exhibit very similar performance. In contrast, Core i7 models from the same generations register more noticeable, albeit small, increases—possibly associated with higher frequencies or optimizations targeted at intensive workloads.

A notable phenomenon is the equivalence in performance between models of different categories across different time periods. The Core i3-9100, for example, is comparable to the Core i5-7600, which, in turn, shows similar performance to the Core i7-4790K (a 2014 high-end model) and the Ryzen 7 1700X. This convergence highlights how quickly innovations in entry-level or mid-range processors can surpass top-tier models from previous generations, evidencing the fast-paced technological evolution in the semiconductor industry.

4.3 Multi-Core Equivalence Chart

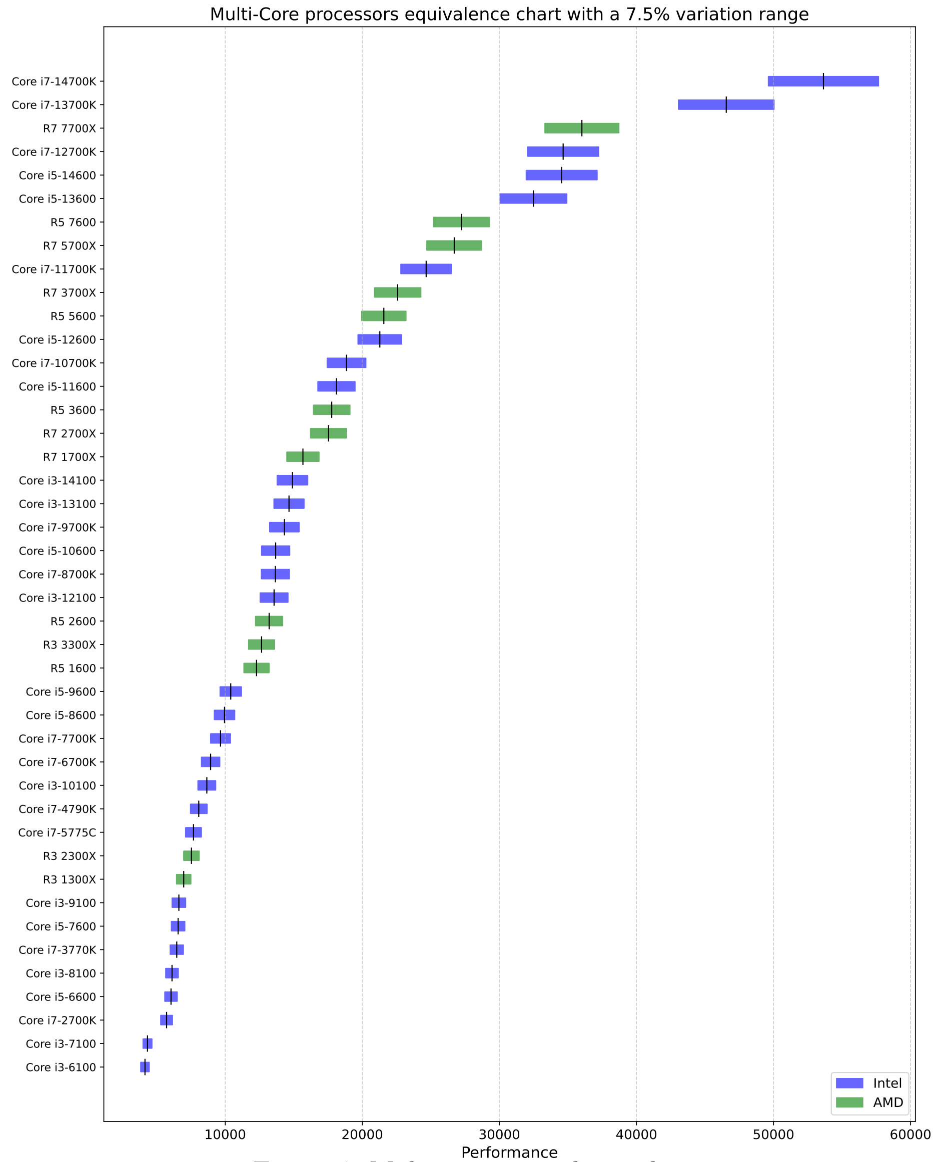

Figure 4: Multi core equivalence chart

When analyzing the evolution of multi-core performance, it is also possible to identify some patterns. An important point to observe is the time it takes for lower-end processors to reach the performance levels of more expensive ones: the Intel Core i7 3770K (high-end from 2013) shows multi-core performance comparable to the Intel Core i5 6600 (mid-range from 2015), which, in turn, is equivalent to the Core i3 8100 (entry-level, 8th generation, 2017). This performance overlap between models of different tiers over a four-year span confirms that entry-level processors can, within a few years, match or surpass premium solutions from previous generations.

A similar trend is observed in more recent models. The Intel Core i7 9700K (high-end from 2018), for instance, demonstrates multi-core performance equivalent to the Core i5 10600 (mid-range from 2020) and the AMD Ryzen 5 2600 (mid-range from 2018). This comparison reinforces the point that, in approximately two years, a high-end processor will have performance very similar to that of a mid-range one, such as the i5 10600. The Intel Core i7 12700K (2021) also maintains equivalence with the Core i5 14600 (2023), once again highlighting this two-year interval.

Of particular note, on the other hand, is the stagnation in multi-core performance of models like the Core i3 6100 (6th generation, 2015) and i3 7100 (7th generation, 2016), which maintained similar scores for several years, reflecting a period of stagnation in advancements for entry-level CPUs. However, this trend is broken with the Core i3 13100 and 14100, which incorporate significant increases in the number of cores and threads.

5 Conclusion

This study characterized the performance evolution of processors from the Intel Sandy Bridge and AMD Zen architectures onward, with the objective of developing a methodology to guide consumer hardware upgrade decisions. The quantitative analysis revealed two distinct growth trends: a linear progression in single-core performance and an exponential increase in multi-core capabilities. These findings demonstrate that, from a general perspective, processor performance has continued to evolve without stagnation. The exponential multi-core growth is primarily driven by a consistent increase in core counts, a strategy that shifts the performance optimization burden from hardware design to parallel software development. A key milestone in this evolution was the paradigm shift introduced with Intel's 12th generation processors (Alder Lake), which incorporated a hybrid architecture of performance and efficiency cores. This design marked a significant change in how processing resources are allocated for different workloads, further accelerating multi-core performance gains.

The development of a equivalence graphics provides a practical framework for consumers, demonstrating that the performance of a high-performance processor, such as a Core i7, tends to be matched or surpassed by an entry-level model within a predictable two to four-year timeframe. Thus, the equivalence graphics is a powerfull tool which make the users avoid unnecessary spending on upgrades that may offer only marginal performance gains while also providing more CPUs to be considered at a specific performance interval.

Consequently, the analysis promotes more conscious and sustainable technological consumption by matching the useful life of components with the customer necessities, thus mitigating the generation of electronic waste. As future work, this analysis could be applied to other benchmarks and computer parts, including metrics for energy efficiency or performance in specific applications.

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